

Groups Sheet 1

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Let G be any group. Show that the identity e is the only element $g \in G$ satisfying the equation $g^2 = g$.

Proof. Take g to be any element of the group satisfying $g^2 = g$. Multiply on the right by g^{-1} to get

$$g \cdot g \cdot g^{-1} = g \cdot g^{-1}$$

$$g \cdot e = e$$

$$g = e$$

□

Let H and K be two subgroups of a group G . Show that the intersection $H \cap K$ is a subgroup of G . Show that the union $H \cup K$ is a subgroup of G if and only if either $H \subseteq K$ or $K \subseteq H$.

Proof. (i) Since we have that H, K are subgroups, they must necessarily contain e and thus their intersection contains e . To check inverses, if $g \in H \cap K$ then since $g^{-1} \in H$ and $g^{-1} \in K$ we have that $g^{-1} \in H \cap K$. For closure, if $a, b \in H \cap K$, then $a, b \in H$ and $a, b \in K$ by the properties of subgroups imply that $ab \in H$ and $ab \in K$. Thus, $ab \in H \cap K$. Since we have all of the properties of a subgroup, we know that $H \cap K$ is a subgroup.

(ii) ($\implies$) Since both contain e we know that the two subsets cannot be disjoint. We prove the contrapositive. Take $h \in H \setminus K$ and $k \in K \setminus H$. Consider hk ; $hk \notin H$ since k is not an element of h . With a similar argument for K , we see that $hk \notin H \cup K$ and as such we do not have closure.

($\impliedby$) If we have that $H \subseteq K$ or that $K \subseteq H$ then $H \cup K$ is equal to K or to H respectively. Since we know that these are subgroups, we are finished.

□

Let $G = \mathbb{R} \setminus \{-1\}$, and let $x * y = x + y + xy$, where xy denotes the usual product of two real numbers. Show that $(G, *, 0)$ is a group. What is the inverse of 2 in this group? Solve the equation $2 * x * 5 = 6$.

Proof. G is a group if it satisfies identity, closure, associativity, and inverses.

(i) We are given that 0 is the identity.

$$0 * 0 = 0 + 0 + (0)(0) = 0.$$

(ii) Given $x, y \in \mathbb{R}$, we know that the usual operations of addition and multiplication will result in a real number. What remains to show is that we never have $x * y = -1$, or rather that

$$-1 = x + y + xy$$

has no real solutions.

$$-1 = x + y + xy$$

$$-1 = x + y(1 + x)$$

$$-(1 + x) = y(1 + x)$$

This equality can only hold if either $x = -1$ or $y = -1$, which they cannot be. Thus we have closure

(iii) Associativity is satisfied when $x * (y * z) = (x * y) * z$. Since real addition is associative:

$$\begin{aligned}
 x * (y * z) &= x * (y + z + yz) \\
 &= x + (y + z + yz) + x(y + z + yz) \\
 &= x + y + z + yz + xy + xz + xyz \\
 &= x + y + xy + z + zx + yz + xyz \\
 &= (x + y + xy) + z + (x + y + xy)z \\
 &= (x + y + xy) * z \\
 &= (x * y) * z.
 \end{aligned}$$

(iv) The inverse for any element $x \in G$ is $x^{-1} = -\frac{x}{1+x}$.

$$\begin{aligned}
 x * x^{-1} &= x + \left(-\frac{x}{1+x}\right) + x\left(-\frac{x}{1+x}\right) \\
 &= \frac{x + x^2}{1+x} - \frac{x}{1+x} - \frac{x^2}{1+x} \\
 &= 0
 \end{aligned}$$

The inverse of 2 is therefore $-\frac{2}{3}$.

To solve $2 * x * 5 = 6$, we determine the inverse elements of 2, 5 and left and right multiply accordingly.

$$\begin{aligned}
 x &= 2^{-1} * 6 * 5^{-1} \\
 &= -\frac{2}{3} * 6 * -\frac{5}{6} \\
 &= -\frac{11}{18}
 \end{aligned}$$

□

Let G be a finite group.

- (a) Let $g \in G$. Show from first principles that there is a positive integer n such that $g^n = e$. (The least such n is called the *order* of g .)

Proof. We can use a simple induction argument to know that for all $k \in \mathbb{N}$, we have that $g^k \in G$. By the pidgeonhole principle, we know that there exists some $i, j \in \mathbb{N}, i > j$ such that $g^i = g^j$. From the properties of G as a group Since we know we can always cancel, we have that $g^{i-j} = e$ where $i - j = n \in \mathbb{Z}^+$. □

- (b) Show from first principles that there is a positive integer N such that $g^N = e$ for all $g \in G$.

Proof. Since for each $g \in G$ there is an integer n that satisfies $g^n = e$, then if we take $N = \text{lcd}(n_1, \dots, n_k)$ for $|G| = k$, then this is the integer such that every $g^N = e$. □

Let S be a finite non-empty set of non-zero complex numbers which is closed under multiplication. Show that S is a subset of the set $\{z \in \mathbb{C} \mid |z| = 1\}$. Show that S is a group with respect to multiplication, and deduce that S is the set of n th roots of unity for some $n \in \mathbb{N}$; that is,

$$S = \{e^{2\pi i k/n} \mid k = 0, 1, \dots, n-1\}.$$

Proof. If $z \in S$ is complex then we can represent it as $z = |z|(\cos \theta + i \sin \theta)$ where $\theta = \arg(z)$. Since S is finite then we must have, for each element, some $n \in \mathbb{N}$ such that $g^n = e$, or in this instance, $g^n = 1 + 0i$. Using De Moivre's theorem, we have that $g^n = |z|^n(\cos n\theta + i \sin n\theta)$. We can see that $|z| = 1$ must be the case. Thus, this set is a subset of $\{z \in \mathbb{C} \mid |z| = 1\}$. □